Lab Experiments 18 -22

Set 18: Product Inventory Management

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
inventory <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300),
  Price = c(20, 15, 18)
)
barplot(inventory$Quantity,
        names.arg = inventory$Product,
        col = "skyblue",
        main = "Quantity of Products in Inventory",
        xlab = "Products",
        ylab = "Quantity Available")
```

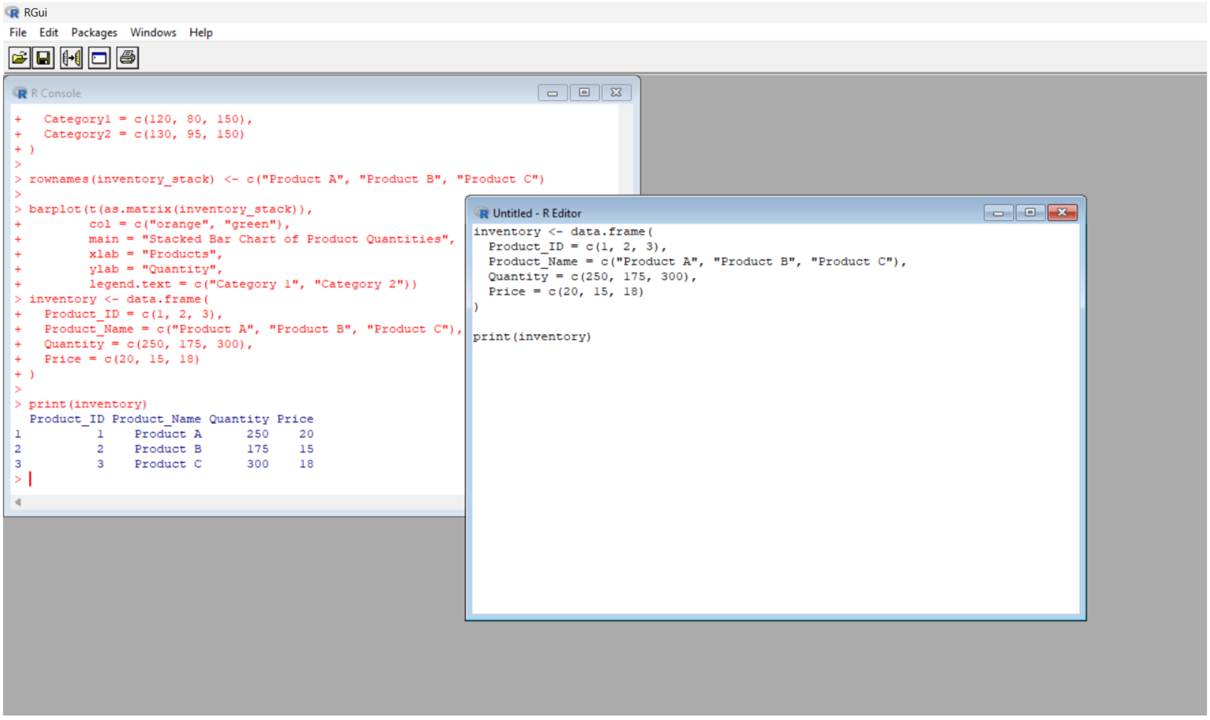
2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
inventory_stack <- data.frame(
  Category1 = c(120, 80, 150),
  Category2 = c(130, 95, 150)
)
rownames(inventory_stack) <- c("Product A", "Product B", "Product C")
barplot(t(as.matrix(inventory_stack)),
        col = c("orange", "green"),
        main = "Stacked Bar Chart of Product Quantities",
        xlab = "Products",
        ylab = "Quantity",
        legend.text = c("Category 1", "Category 2"))
```

3. Build a table to show the inventory data for each product. Label the table elements.

```
inventory <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300),
```

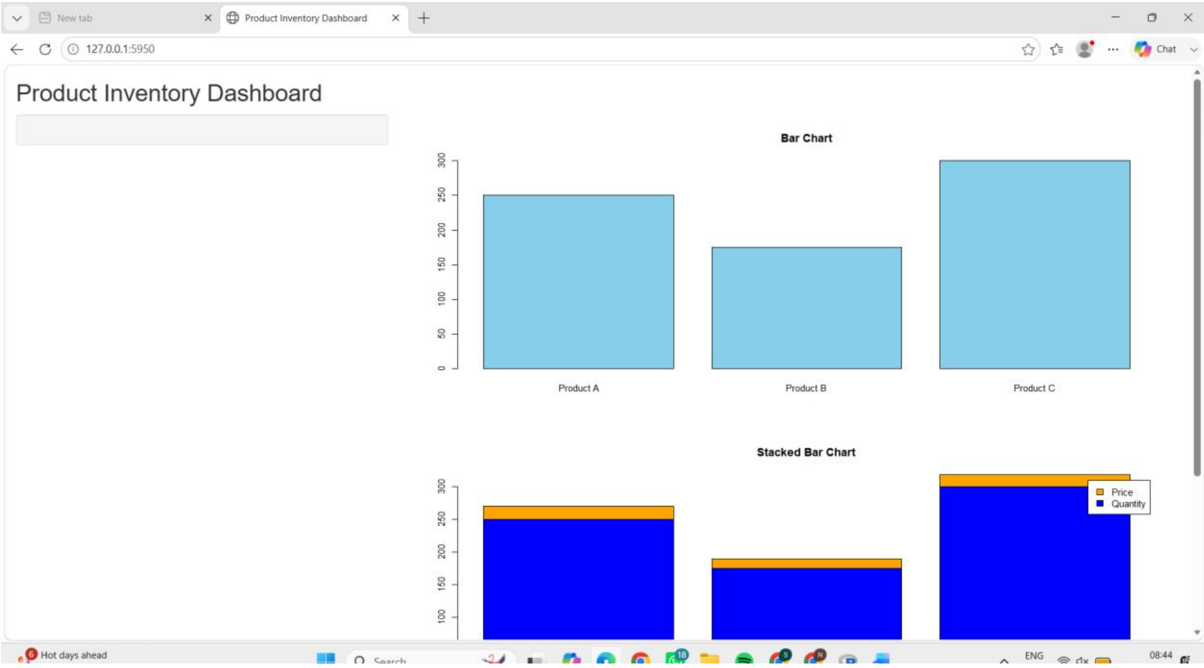
```
  Price = c(20, 15, 18)
)
print(inventory)
```

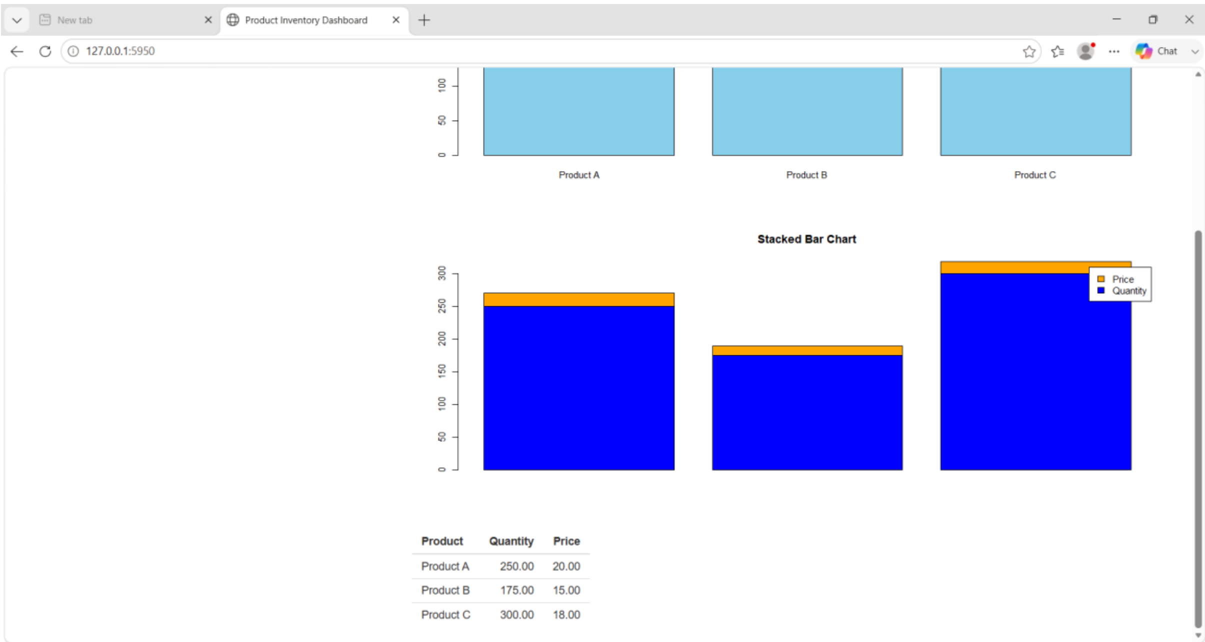


4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

```
install.packages("shiny")
library(shiny)
inventory <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300),
  Price = c(20, 15, 18)
)
ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(),
```

```r
    mainPanel(
      plotOutput("barplot"),
      plotOutput("stackplot"),
      tableOutput("table")
    )
  )
)
server <- function(input, output) {
  output$barplot <- renderPlot({
    barplot(inventory$Quantity,
            names.arg = inventory$Product,
            col = "skyblue",
            main = "Bar Chart")
  })
  output$stackplot <- renderPlot({
    data <- rbind(inventory$Quantity, inventory$Price)
    barplot(data,
            beside = FALSE,
            col = c("blue", "orange"),
            legend = c("Quantity", "Price"),
            main = "Stacked Bar Chart")
  })
  output$table <- renderTable({
    inventory
  })
}
shinyApp(ui = ui, server = server)
```



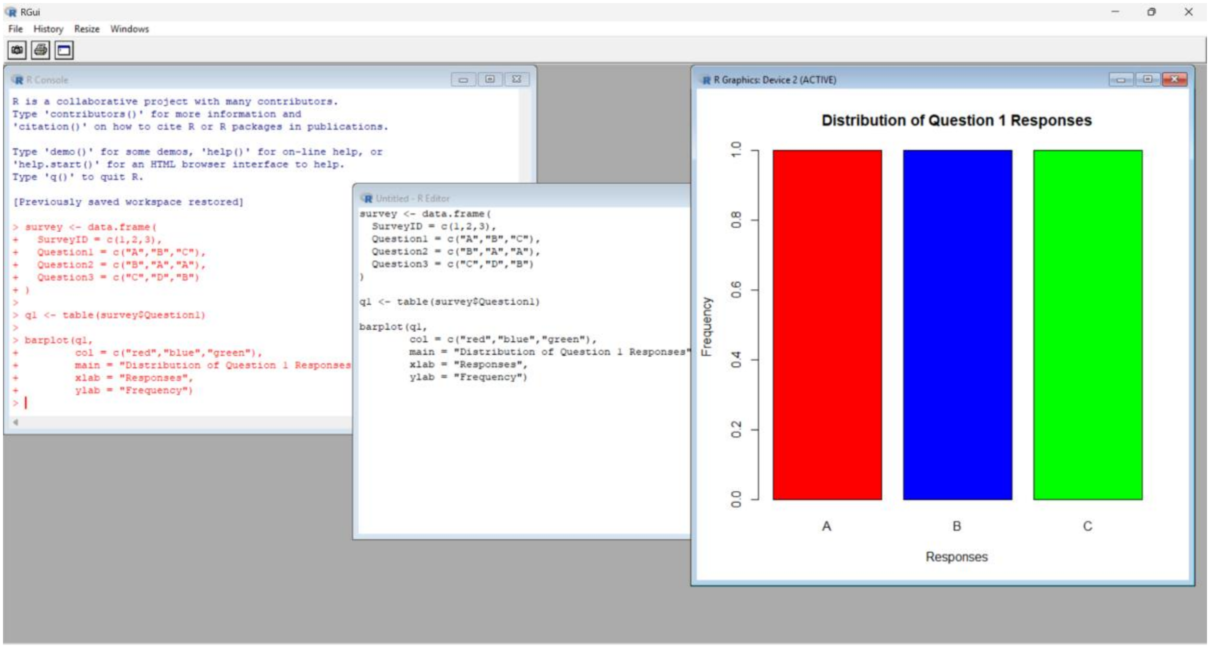


Product	Quantity	Price
Product A	250.00	20.00
Product B	175.00	15.00
Product C	300.00	18.00

Set 19: Survey Responses Analysis

19.1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)


q1 <- table(survey$Question1)


barplot(q1,
        col = c("red","blue","green"),
        main = "Distribution of Question 1 Responses",
        xlab = "Responses",
        ylab = "Frequency")
```

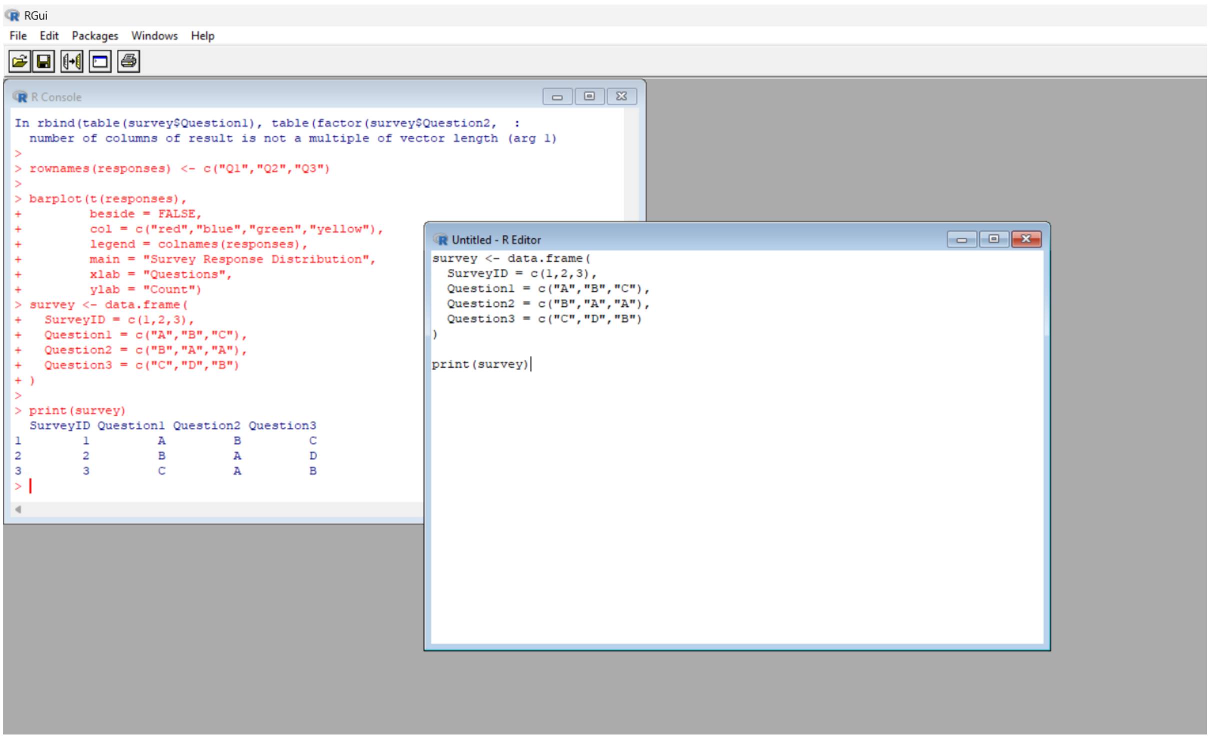


19.2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
responses <- rbind(
  table(survey$Question1),
  table(factor(survey$Question2, levels=c("A","B","C","D"))),
  table(factor(survey$Question3, levels=c("A","B","C","D")))
)


rownames(responses) <- c("Q1","Q2","Q3")


barplot(t(responses),
        beside = FALSE,
        col = c("red","blue","green","yellow"),
        legend = colnames(responses),
        main = "Survey Response Distribution",
        xlab = "Questions",
        ylab = "Count")
```

19.3. Build a table to show the survey response data for each survey. Label the table elements.

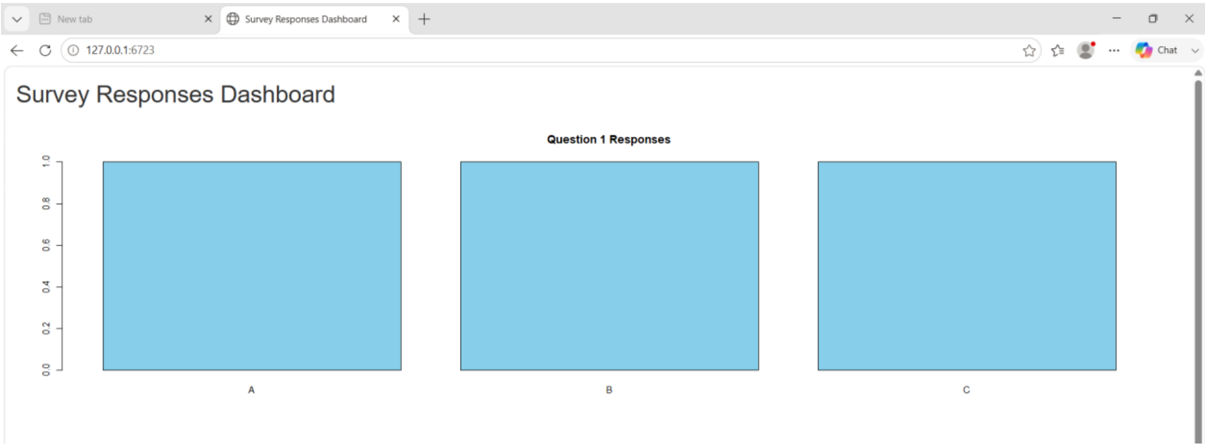
```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)


print(survey)
```

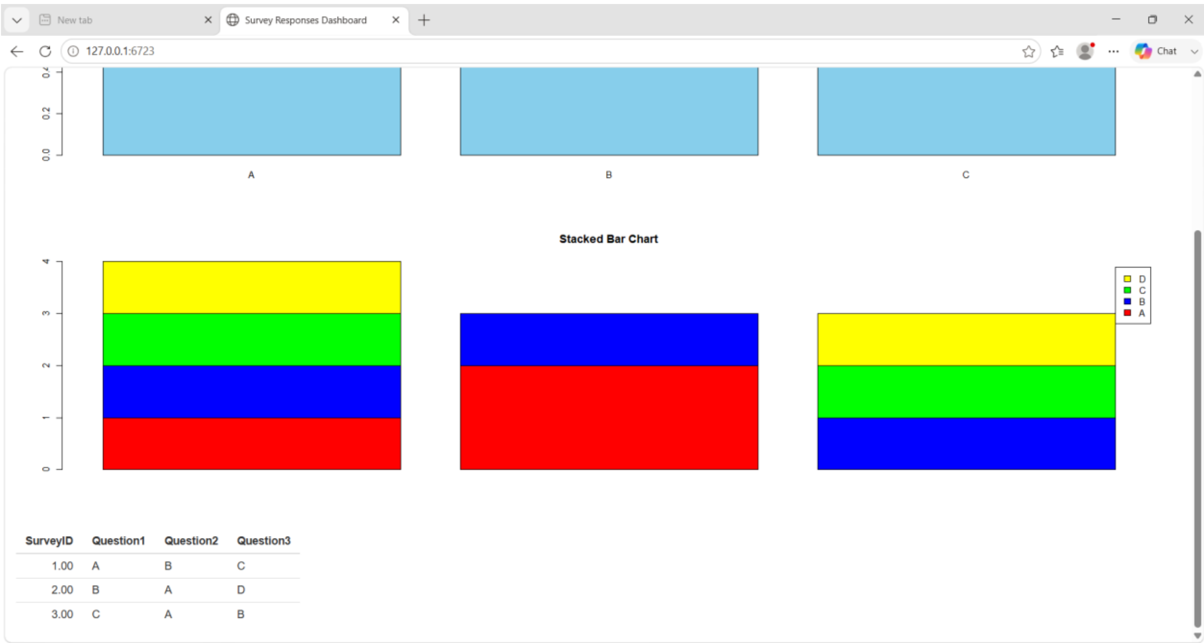


19.4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

```
install.packages("shiny")
library(shiny)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  plotOutput("bar1"),
  plotOutput("bar2"),
  tableOutput("table")
)
server <- function(input, output){
  output$bar1 <- renderPlot({
    barplot(table(survey$Question1),
            col="skyblue",
            main="Question 1 Responses")
  })
  output$bar2 <- renderPlot({
    data <- rbind(
      table(survey$Question1),
      table(factor(survey$Question2, levels=c("A","B","C","D"))),
      table(factor(survey$Question3, levels=c("A","B","C","D")))
    )
```

```
  barplot(t(data),
          beside=FALSE,
          col=c("red","blue","green","yellow"),
          legend=colnames(data),
          main="Stacked Bar Chart")
 })
 output$table <- renderTable({
  survey
 })
}
shinyApp(ui, server)
```





Set 20: Stock Analysis

20.1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

```
date <- as.Date(c("2023-01-01","2023-01-02","2023-01-03"))

stockA <- c(100,105,110)

stockB <- c(150,152,148)

stockC <- c(120,118,122)

plot(date, stockA, type="o", col="blue",

   ylim=c(90,160),

   xlab="Date",

   ylab="Stock Price",

   main="Stock Prices")

lines(date, stockB, type="o", col="red")

lines(date, stockC, type="o", col="green")

legend("topleft",

     legend=c("Stock A","Stock B","Stock C"),

     col=c("blue","red","green"),

     lty=1,

     pch=1)
```

20.2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

change <- c(0, (105-100)/100*100, (110-105)/105*100)

barplot(change,

names.arg=c("Jan 1","Jan 2","Jan 3"),

col="orange",

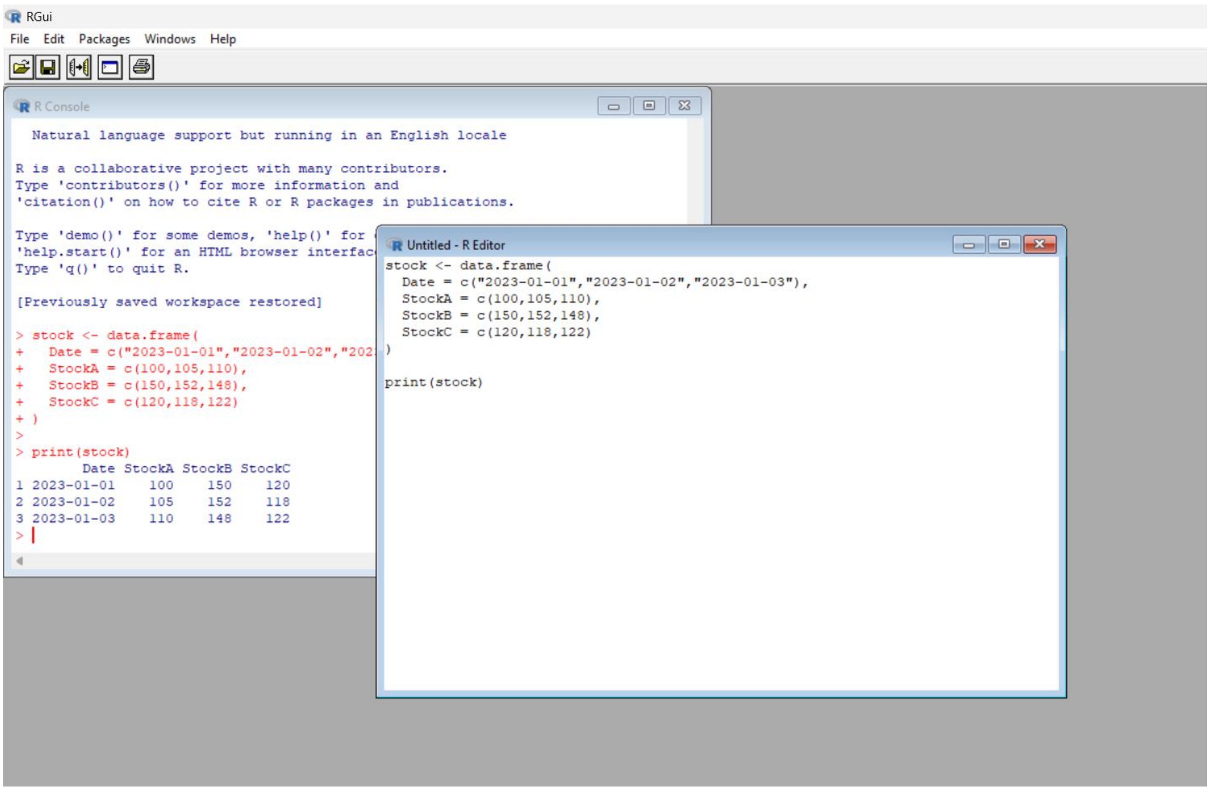
main="Daily % Change - Stock A",

xlab="Date",

ylab="Percentage Change")

20.3. Build a table to display the stock price data for each company over the given period. Label the table elements.

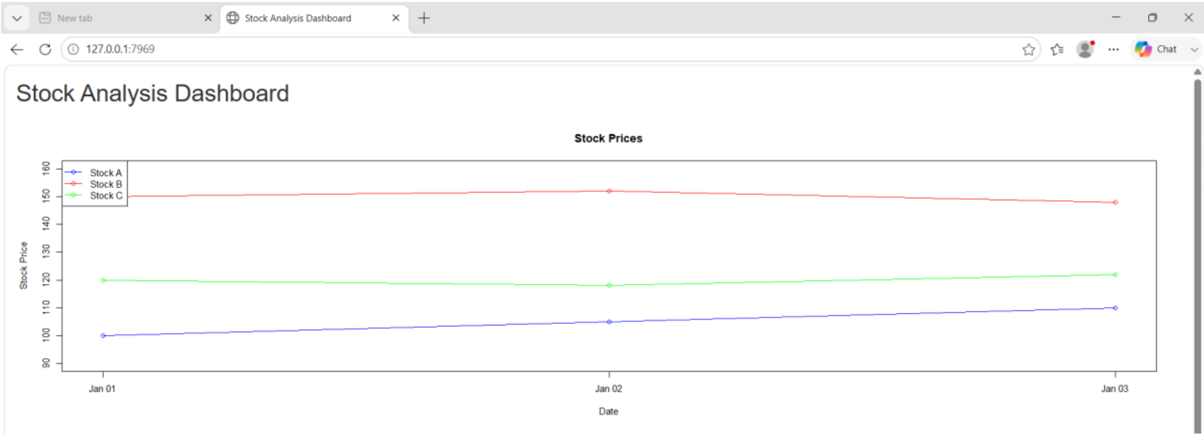
```
stock <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03"),
  StockA = c(100,105,110),
  StockB = c(150,152,148),
  StockC = c(120,118,122)
)


print(stock)
```

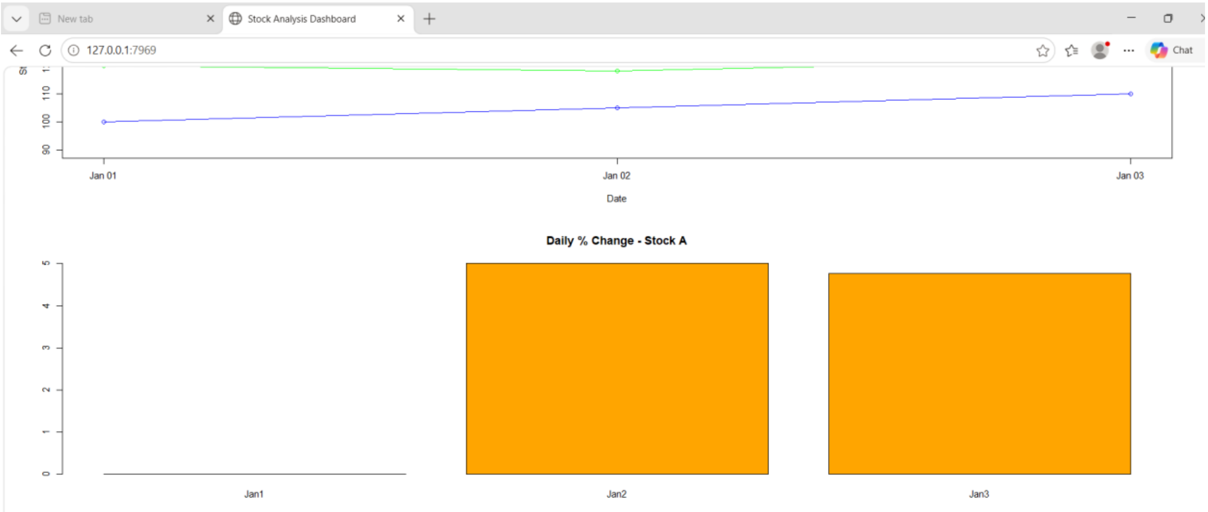


20.4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

```
install.packages("shiny")


library(shiny)


stock <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03"),
  StockA = c(100,105,110),
  StockB = c(150,152,148),
  StockC = c(120,118,122)
)


ui <- fluidPage(
  titlePanel("Stock Analysis Dashboard"),
  plotOutput("line"),
  plotOutput("bar"),
  tableOutput("table")
)


server <- function(input, output){


  output$line <- renderPlot({
    date <- as.Date(stock$Date)


    plot(date, stock$StockA, type="o", col="blue",
         ylim=c(90,160),
         xlab="Date",
         ylab="Stock Price",
```

```
      main="Stock Prices")

    lines(date, stock$StockB, type="o", col="red")
    lines(date, stock$StockC, type="o", col="green")

    legend("topleft",
           legend=c("Stock A","Stock B","Stock C"),
           col=c("blue","red","green"),
           lty=1,
           pch=1)
  })

  output$bar <- renderPlot({
    change <- c(0,
                (105-100)/100*100,
                (110-105)/105*100)

    barplot(change,
            names.arg=c("Jan1","Jan2","Jan3"),
            col="orange",
            main="Daily % Change - Stock A")
  })

  output$table <- renderTable({
    stock
  })
}

shinyApp(ui, server)
```



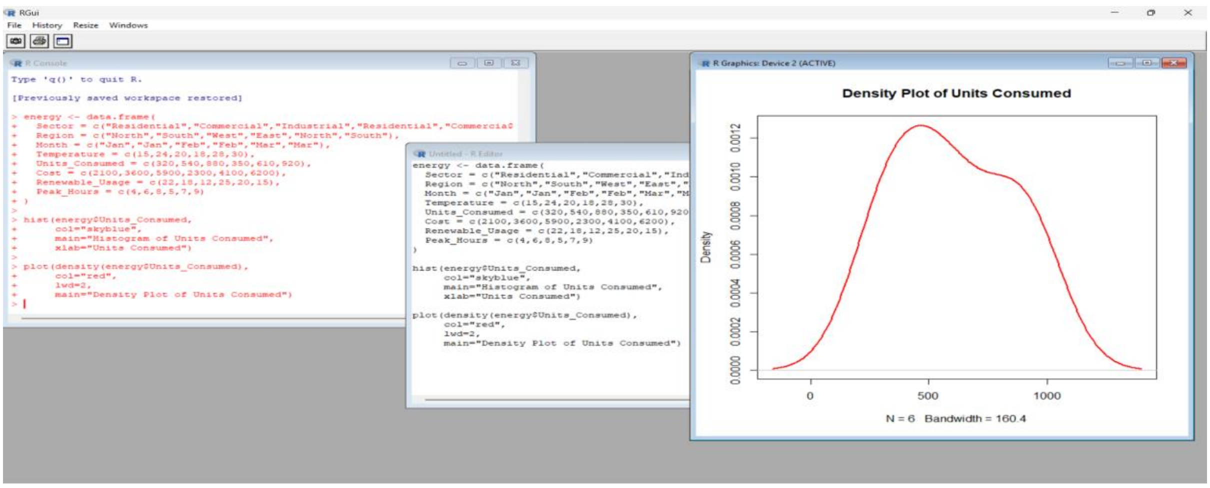


Date	StockA	StockB	StockC
2023-01-01	100.00	150.00	120.00
2023-01-02	105.00	152.00	118.00
2023-01-03	110.00	148.00	122.00

Set 21

21.1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

```
energy <- data.frame(
  Sector =
c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region = c("North","South","West","East","North","South"),
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature = c(15,24,20,18,28,30),
  Units_Consumed = c(320,540,880,350,610,920),
  Cost = c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage = c(22,18,12,25,20,15),
  Peak_Hours = c(4,6,8,5,7,9)
)
hist(energy$Units_Consumed,
     col="skyblue",
     main="Histogram of Units Consumed",
     xlab="Units Consumed")
plot(density(energy$Units_Consumed),
     col="red",
     lwd=2,
     main="Density Plot of Units Consumed")
```



21.2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

plot(energy$Temperature,

energy$Units_Consumed,

cex=energy$Peak_Hours/2,

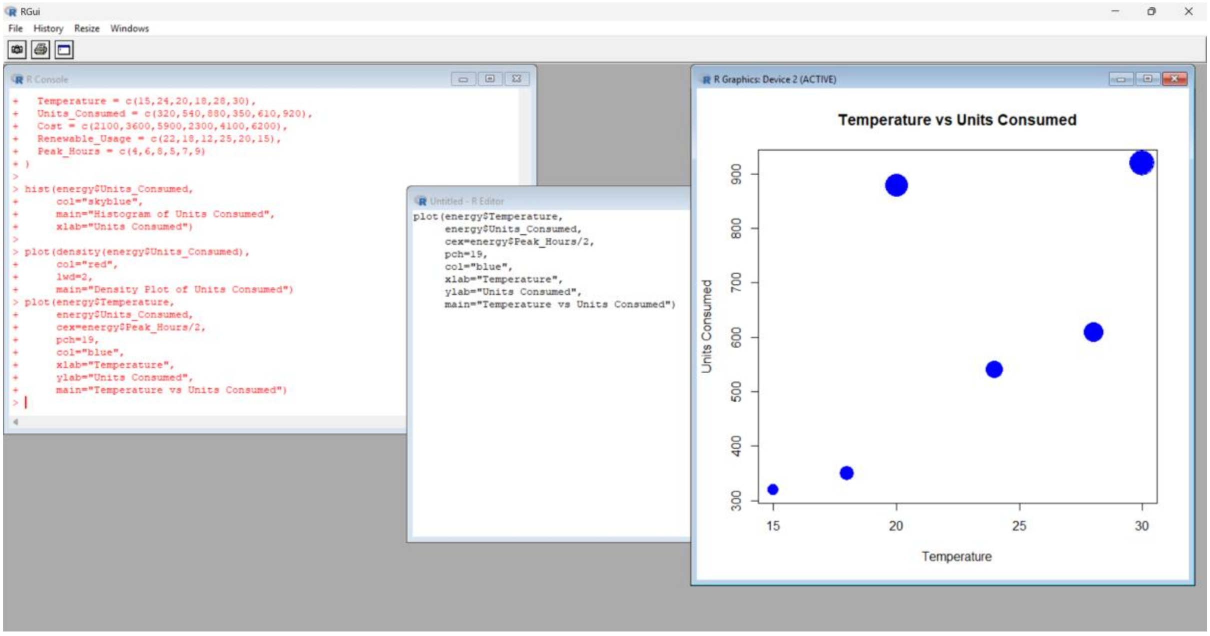
pch=19,

col="blue",

xlab="Temperature",

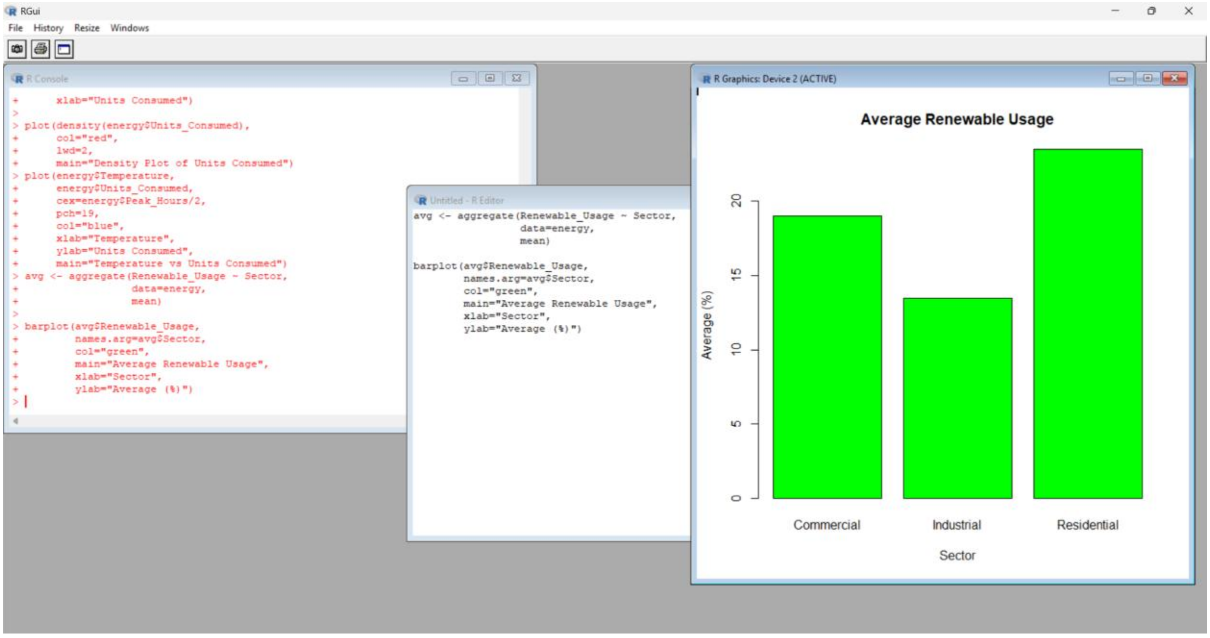
ylab="Units Consumed",

main="Temperature vs Units Consumed")



21.3. Using R, calculate average Renewable Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

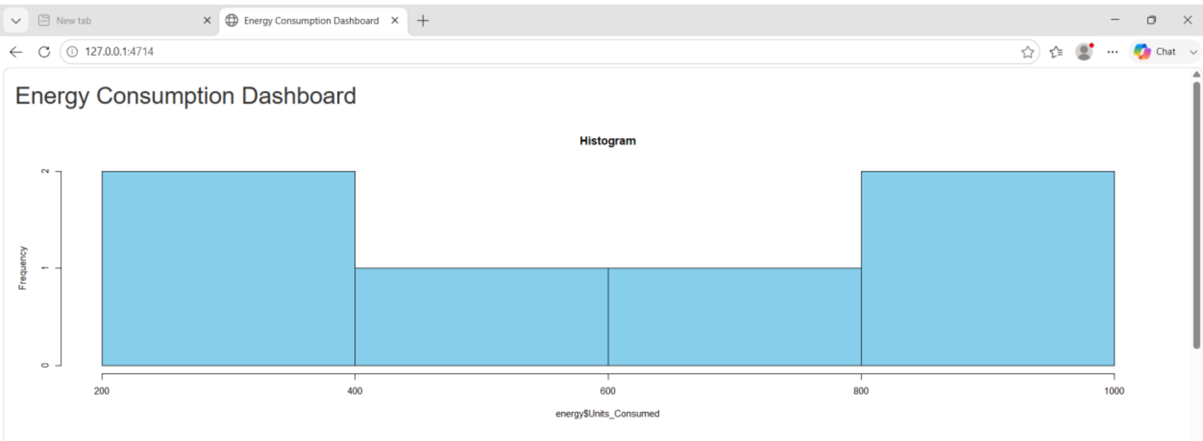
```
avg <- aggregate(Renewable_Usage ~ Sector,
                 data=energy,
                 mean)


barplot(avg$Renewable_Usage,
        names.arg=avg$Sector,
        col="green",
        main="Average Renewable Usage",
        xlab="Sector",
        ylab="Average (%)")
```

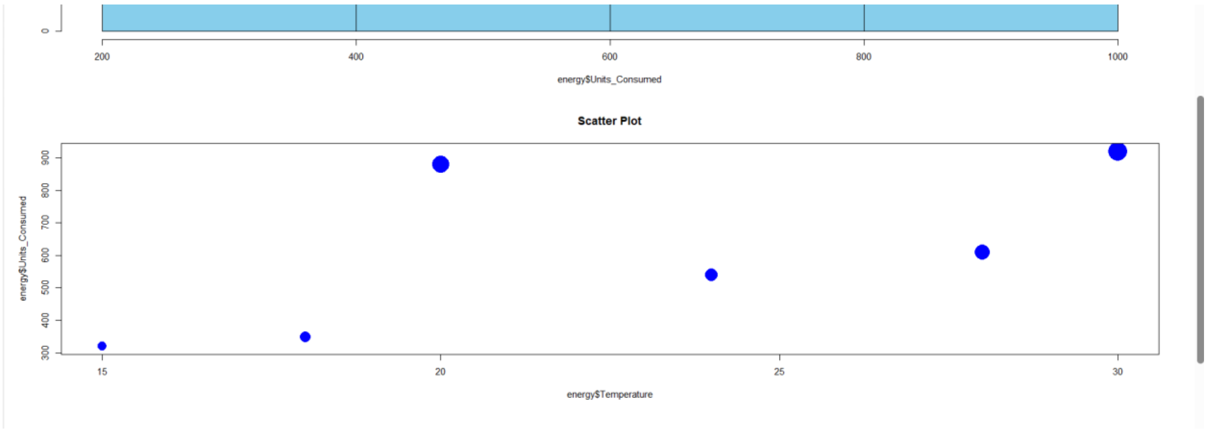


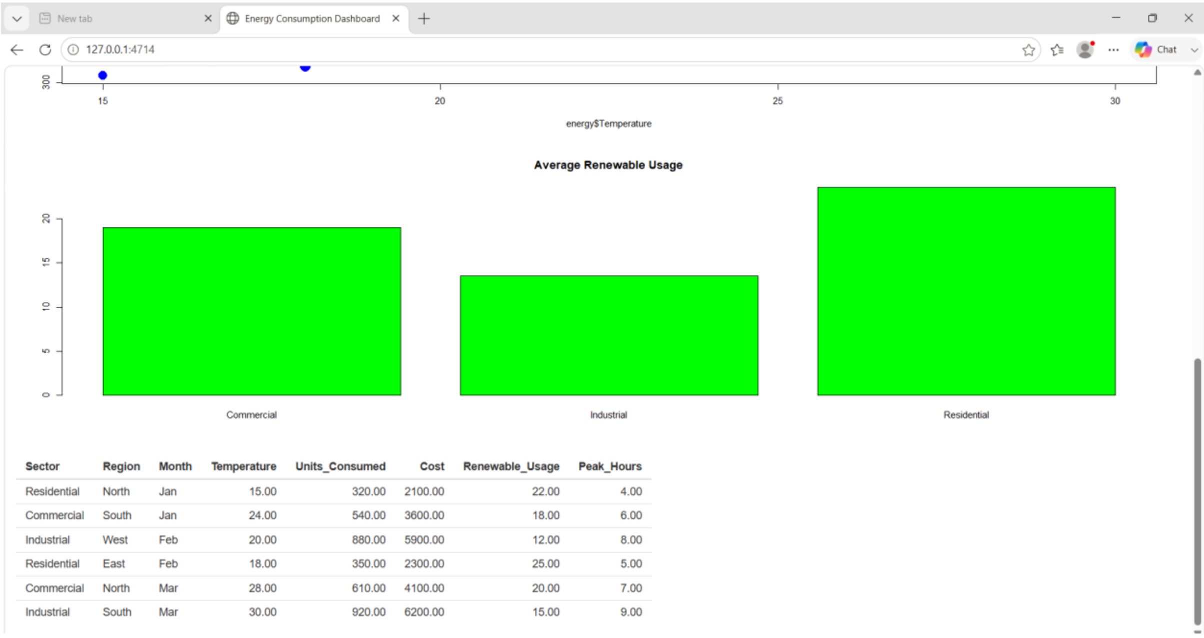
21.4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

```
install.packages("shiny")

library(shiny)


ui <- fluidPage(

  titlePanel("Energy Consumption Dashboard"),


  plotOutput("hist"),

  plotOutput("scatter"),

  plotOutput("bar"),

  tableOutput("table")

)


server <- function(input, output){


  output$hist <- renderPlot({

    hist(energy$Units_Consumed,

         col="skyblue",

         main="Histogram")

  })


  output$scatter <- renderPlot({

    plot(energy$Temperature,

         energy$Units_Consumed,

         cex=energy$Peak_Hours/2,

         pch=19,

         col="blue",

         main="Scatter Plot")
```

```
  })

  output$bar <- renderPlot({
    avg <- aggregate(Renewable_Usage ~ Sector,
                     data=energy,
                     mean)

    barplot(avg$Renewable_Usage,
            names.arg=avg$Sector,
            col="green",
            main="Average Renewable Usage")
  })

  output$table <- renderTable({
    energy
  })
}

shinyApp(ui, server)
```







Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage	Peak_Hours
Residential	North	Jan	15.00	320.00	2100.00	22.00	4.00
Commercial	South	Jan	24.00	540.00	3600.00	18.00	6.00
Industrial	West	Feb	20.00	880.00	5900.00	12.00	8.00
Residential	East	Feb	18.00	350.00	2300.00	25.00	5.00
Commercial	North	Mar	28.00	610.00	4100.00	20.00	7.00
Industrial	South	Mar	30.00	920.00	6200.00	15.00	9.00

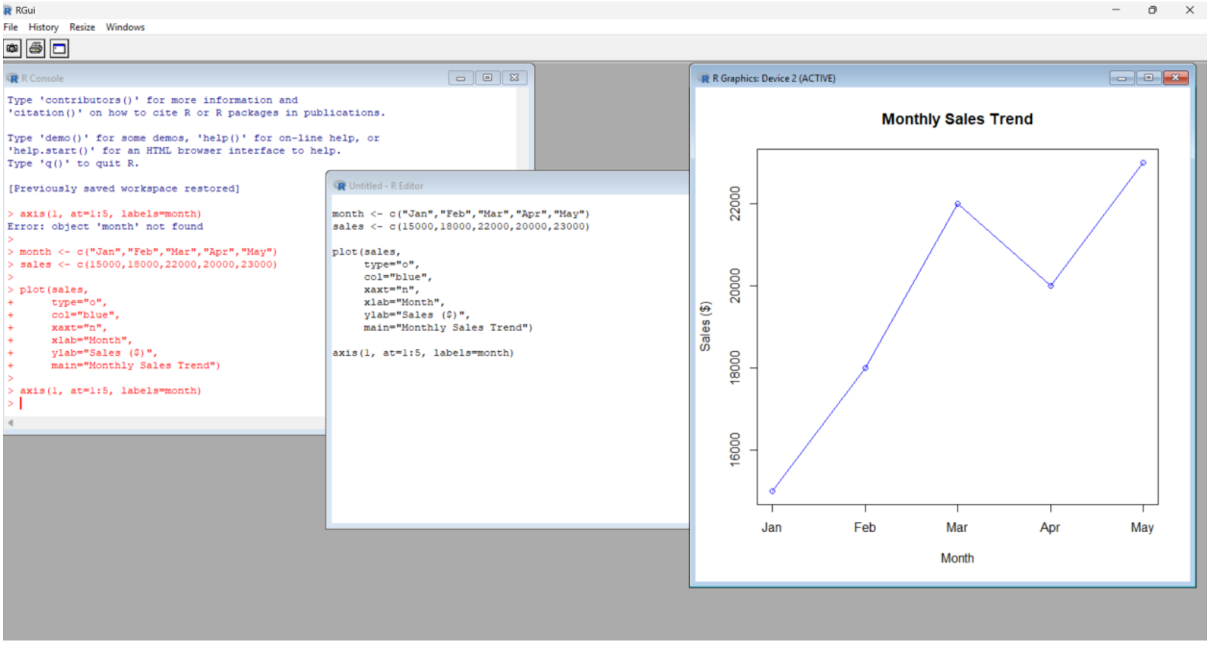
Set 22: Time Series Analysis

22.1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

```
month <- c("Jan","Feb","Mar","Apr","May")

sales <- c(15000,18000,22000,20000,23000)

plot(sales,

    type="o",

    col="blue",

    xaxt="n",

    xlab="Month",

    ylab="Sales ($)",
```

main="Monthly Sales Trend")

axis(1, at=1:5, labels=month)



22.2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

budget <- c(5000,6000,7000,6500,7500)

plot(budget,

sales,

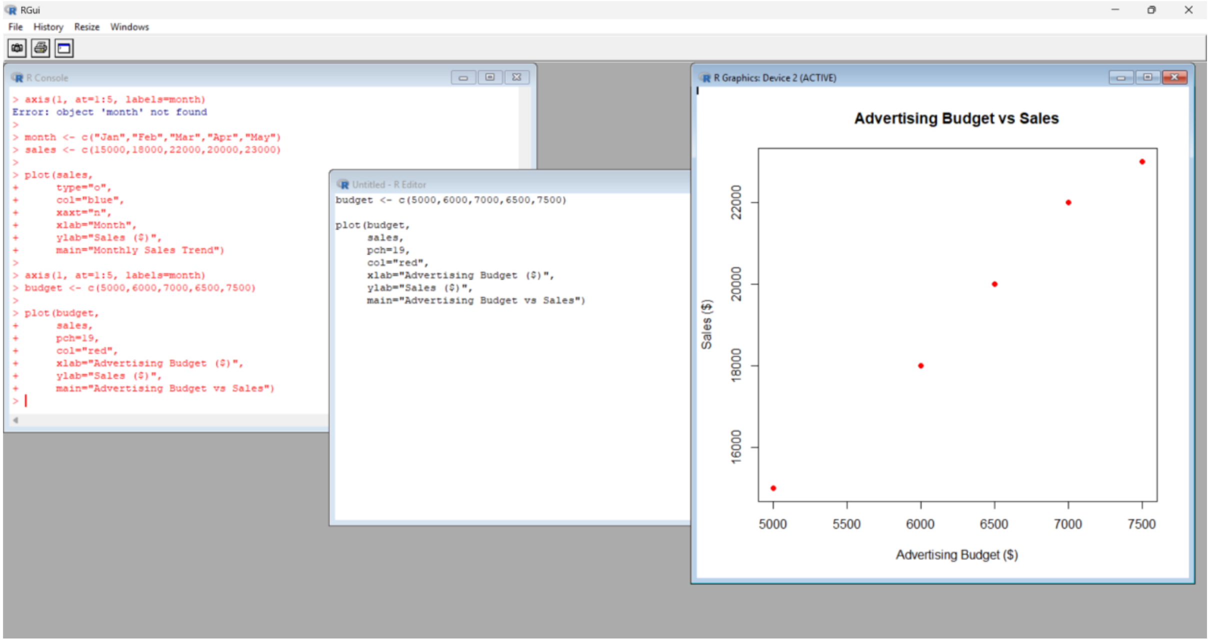
pch=19,

col="red",

xlab="Advertising Budget ($)",

ylab="Sales ($)",

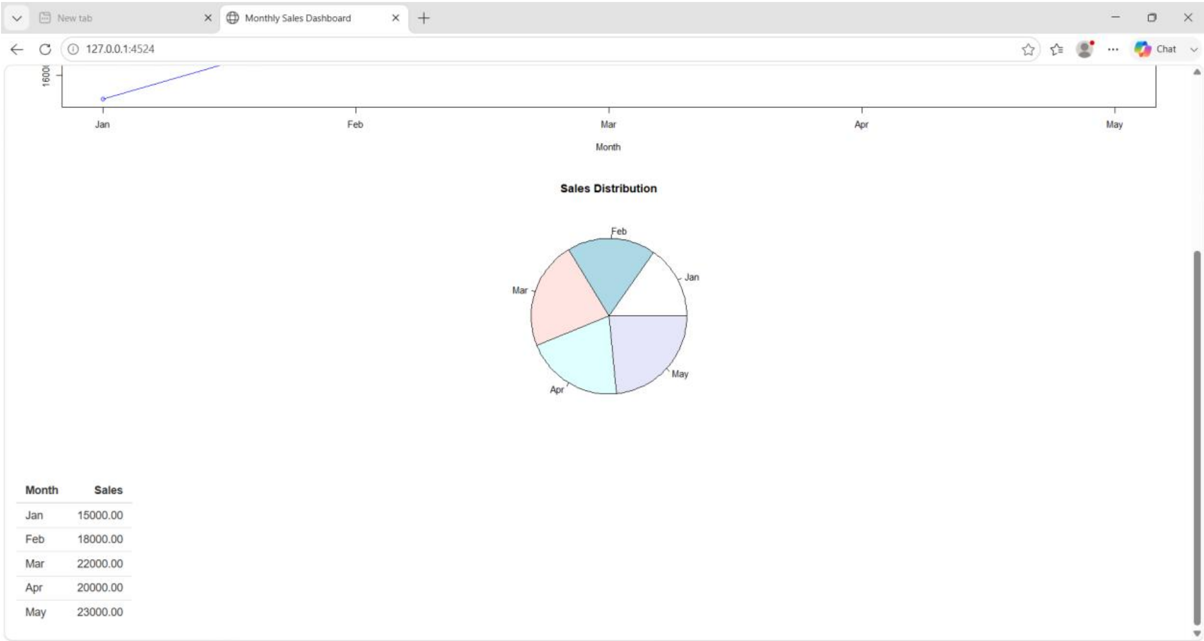
main="Advertising Budget vs Sales")



22.3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

```
install.packages("shiny")

library(shiny)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  plotOutput("line"),

  plotOutput("pie"),

  tableOutput("table")

)

server <- function(input, output){

  output$line <- renderPlot({

    plot(sales,

         type="o",

         col="blue",

         xaxt="n",

         xlab="Month",

         ylab="Sales ($)",
```

```
        main="Monthly Sales Trend")
   axis(1, at=1:5, labels=month)
  })
  output$pie <- renderPlot({
   pie(sales,
       labels=month,
       main="Sales Distribution")
  })
  output$table <- renderTable({
   data.frame(Month=month, Sales=sales)
  })
}
shinyApp(ui, server)
```



22.4. In R, create an autocorrelation plot to identify seasonality in the time series data.

```
sales_ts <- ts(c(15000,18000,22000,20000,23000))

acf(sales_ts,

    main="Autocorrelation Plot of Monthly Sales")
```